Multi-Agent AI Trading System

Updated Integrated Plan for Indian Market (USD/INR + NSE Stocks & Indices) Version 1.3.5 | February 2026

0. Document Control

Purpose: maintain versioning, accountability, and auditability. Acceptance Criteria: - Single owner and reviewer list are defined. - Cross-functional pod ownership is defined for each asset cluster and artifact (primary owner + reviewer). - Versioning and changelog are maintained in this document. - Each model and data artifact is traceable to a plan version. - Version 1.2 change set captures dual-loop execution, sentiment cache decoupling, crisis override voting, and impossible-scenario stress tests. - Version 1.2.1 applies minimum-safe scope: ship dual-loop, sentiment cache, basic crisis mode, and impossible stress testing; defer aggressive alien-state automation, snapback smoothing tuning, and full winner-takes-all automation. - Version 1.3.0 adds hardware-acceleration track, market-making readiness module, research-to-production throughput KPIs, low-latency ML release gates, pod operating model, data-scale roadmap and replay, and change-outcome accountability reviews. - Version 1.3.1 tightens execution latency targets (5 to 8 ms stretch with enforced degrade cap), adds real-time impact and attribution controls, introduces mandatory shadow A/B for major FX-gold changes, and formalizes optional online RL micro-updates under strict safety gates. - Version 1.3.2 removes speculative uplift language, adds explicit no-assumed-speedup policy for Rust/C++/FPGA choices, and introduces phase-by-phase go/no-go benchmark gates. - Version 1.3.3 added evidence-first rigor guidance and prioritization for IMC-inspired tech adoption. - Version 1.3.4 folds rigor guidance directly into MLOps, rollout, and phase-gate sections and removes the standalone memo section. - Version 1.3.5 operationalizes release governance with mandatory evidence-first change-request and CI benchmark checklist templates.

1. Executive Summary

This updated plan refines the existing multi-agent trading system for Indian markets by adding concrete governance, execution, and validation controls. It preserves the core multi-agent architecture while tightening scope, compliance, data integrity, and rollout safety. The plan incorporates a dedicated Sentiment Agent in Phase 2 using a fine-tuned FinBERT model to improve event-driven robustness. Deployment and promotion decisions are governed by explicit benchmark gates in Section 16 and Appendix A rather than assumed performance uplifts. Version 1.3.5 operationalizes those rules through mandatory governance templates in promotion workflows.

2. Scope, Objectives, and Trading Constraints

Defines the trading horizon, asset universe, allowed actions, and performance targets. Acceptance Criteria: - Primary trading frequency is Hourly and explicitly documented. - Daily timeframe is used only for confirmation and regime context filtering. - Data sampling frequency is locked to the primary horizon to avoid model-data mismatch. - Asset universe is explicit and versioned (NSE equities, indices, USD/INR, F&O). - Universe selection policy includes numeric filters: Avg Daily Turnover >= INR 10 Cr (6-month average), Impact Cost <= 0.50 percent, Free float >= 20 percent (per NSE IWF methodology), Price >= INR 50, Listing age >= 12 months, and no stocks with > 5 circuit hits in the last 3 months. - Universe review and rebalance occurs quarterly via the Preprocessing Agent. - Allowed actions are specified per product (shorting, leverage, lot size, margin, order types). - Performance objectives include return, risk, drawdown, capacity, and turnover targets. - Target performance metrics are stated as goals, not guarantees, with confidence intervals.

3. Regulatory and Broker Compliance

Defines compliance requirements and audit trails for SEBI and broker rules. Acceptance Criteria: - Compliance checklist exists for SEBI and broker rules. - Pre-trade checks include position limits, margin rules, and circuit breaker rules. - Audit trail includes order intent, execution, cancellation, and model version used. - Static Mumbai IP whitelisting and broker access controls are documented as production-only requirements. - Regulatory reporting automation and retention policy are documented. - Algorithm approval or notification requirements and periodic audit obligations are documented where applicable. - Data usage policy explicitly prohibits unpublished or embargoed information; only publicly released or contractually licensed feeds are permitted.

4. System Architecture and Interfaces

Defines module boundaries, data flows, APIs, and latency budgets. Acceptance Criteria: - Each module has defined inputs, outputs, and failure modes. - Data flow diagram exists with transport protocols and schemas. - Latency budgets are documented per stage (ingest, features, inference, execution). - Stage SLAs include p50, p95, and p99 targets with hard caps and automatic mode degradation on breach. - Architecture enforces dual-loop separation: Fast Loop (execution critical) and Slow Loop (context and policy updates). - Fast Loop target is p99 <= 8 ms as a stretch objective, with enforced degrade or bypass if p99 exceeds 10 ms, and processes only technical microstructure, order-imbalance features, and cache reads. - Slow Loop supports heavier inference (100 to 500 ms typical) and is asynchronous, never blocking order placement. - Volatility bypass thresholds use hysteresis and cooldown windows to prevent rapid mode flapping. - Deployment topology (Mumbai-based low-latency VPS) is specified. - Minimum-safe release profile is explicit: advanced overrides are disabled by default and only basic crisis-weighted controls are active. - ML production rulebook is explicit: no heavy inference in Fast Loop, student-policy inference only in the execution path, and teacher policy execution stays offline or Slow Loop only. - Model artifact promotion includes deterministic runtime and compression checks; p99 and p99.9 tail-latency budgets are mandatory release gates. - Hardware acceleration policy is explicit: FPGA is permitted only for execution-path critical components with measured latency benefit; non-critical components remain CPU/GPU. - Critical path implementation policy allows Rust/C++ microservices for latency-sensitive functions (for example order-book parsing and routing adapters) while orchestration remains language-agnostic. - No language or hardware path is assumed to deliver fixed speedup multipliers; Rust/C++/FPGA promotion requires measured p99 and p99.9 improvement versus baseline plus operational parity on correctness, observability, and recovery drills.

5. Phase 1: Data Orchestration Layer

Specialized agents for acquisition and preprocessing.

5.1 NSE Sentinel Agent

Real-time and historical market data acquisition from NSE sources. Acceptance Criteria: - Data sources include nsepython, jugad-data, and official NSE APIs or their replacements. - Redundancy exists for core price, volume, and corporate action feeds. - Timestamp validation and clock sync are enforced across all streams. - Official API or exchange/broker endpoints are the primary source, with web scraping restricted to fallback mode. - Signal provenance is tagged per event (primary API vs fallback scraper) and exposed to downstream risk controls. - Primary data-feed failure triggers automatic reduce-only or close-only mode until integrity checks pass.

5.2 Macro Monitor Agent

Indian macro indicators and cross-asset signals. Acceptance Criteria: - Macro indicators list is defined and versioned. - Indicators include WPI, CPI, IIP, FII/DII flows, FX reserves, RBI bulletins, and India-US 10Y spread. - Global proxies such as Brent and DXY are explicitly justified as India-relevant or removed. - Data quality checks cover missingness, outliers, and latency.

5.3 Preprocessing Agent

Normalization, noise filtering, corporate actions, and feature readiness. Acceptance Criteria: - Feature engineering is deterministic and reproducible. - Rolling window normalization and directional change thresholds are implemented. - Corporate action adjustments are validated against a reference source. - Leakage tests confirm strict time alignment and lagging. - Feature approval process is defined (proposal, offline evaluation, shadow testing, promotion).

5.4 Textual Data Agent

Text collection for sentiment analysis and event detection. Acceptance Criteria: - Source list includes NSE news, Economic Times, RBI reports, and earnings transcripts. - Social data collection from X is documented with keyword and semantic search rules. - PDF extraction pipeline is validated with spot checks. - Code-mixed English and Hinglish handling strategy is documented.

5.5 Data Scale Roadmap and Replay

Data platform growth and reproducibility posture. Acceptance Criteria: - Data processing roadmap defines current and target daily processing volumes (GB to TB) with quarterly checkpoints. - Dataset tiering is explicit: hot (in-memory/cache), warm (low-latency store), and cold (archive) with retention policies. - Replay framework can reconstruct any trading day from raw feed payloads through feature artifacts with deterministic identifiers. - Replay supports both event-time and wall-clock playback for strategy diagnostics and latency forensics.

6. Phase 2: Analyst Board (Modeling)

Specialized models for forecasting, regime detection, sentiment, and consensus.

6.1 Technical Agent

Time-series modeling for price prediction and action thresholds. Acceptance Criteria: - Model families include ARIMA-LSTM hybrids, 2D CNNs, and GARCH quantiles for VaR and ES. - Model targets and metrics are specified for each family. - Baselines and ablation tests are implemented.

6.2 Regime Agent

Regime detection and transition modeling. Acceptance Criteria: - PEARL or an equivalent meta-learning model is specified. - Regime definitions include RBI inner and outer band zones. - Transition criteria cover event-driven shifts and volatility regime changes. - Calibration tests and stability checks are documented. - Regime training and stress validation include structural breaks (2008 crisis, 2013 taper tantrum, COVID window, and RBI policy discontinuities). - Novelty or out-of-distribution detection uses explicit statistical distance thresholds with an alien-state flag. - Alien-state detection triggers staged de-risking (full risk, reduced risk, neutral or cash) with documented escalation.

6.3 Sentiment Agent

FinBERT-based sentiment modeling for Indian markets. Acceptance Criteria: - Base model is ProsusAI/finbert or equivalent with documented justification. - Fine-tuning datasets include Indian sources such as harixn/indian_news_sentiment and SEntFiN. - Bayesian priors or regularization are defined for stability. - Daily sentiment aggregates form a threshold variable (z_t) used in regime logic. - Precision and recall thresholds are defined by class, with timestamp alignment tests. - Robustness controls include spam and adversarial text filtering, source deduplication, and noise handling. - Dual-speed sentiment design is defined: intraday fast lane (lightweight model plus keyword rules) and offline slow lane (deep model, nightly updates). - Sentiment runs on an independent timeline and writes outputs to fast cache (e.g., Redis) with timestamp, confidence, and TTL. - Fast Loop consumes the latest cached sentiment via O(1) memory reads and never waits for fresh NLP inference. - Intraday sentiment path targets <= 100 ms score generation from headline arrival under normal load, outside the execution-critical path. - Sentiment freshness TTL and event-priority routing are documented to prevent stale text overriding fresh market signals. - Cache decision policy is deterministic: fresh -> use, stale -> downweight, expired -> ignore. - Cache read failure triggers technical-only reduced-risk execution mode until cache health recovers. - Pump-and-dump and slang-scam detection for code-mixed text is required; manipulative hype spikes can force sentiment downgrade to neutral or protective.

6.4 Consensus Agent

Aggregates signals using LSTAR or ESTAR with Bayesian estimation. Acceptance Criteria: - Logistic and exponential transition functions are specified. - Bayesian estimation of smoothness and location parameters is documented. - Transition function includes volatility, macro differentials, RBI signals, and sentiment quantiles. - Consensus stability is tested during high-volatility windows. - Default mode uses weighted consensus with explicit safety bias toward protective signals. - Basic crisis mode uses crisis-weighted routing with capped dominance (max 60 to 70 percent for the crisis agent), not full winner-takes-all. - Agent divergence (fundamental disagreement between agents) is modeled as a dedicated regime signal that triggers staged risk reduction or temporary freeze. - Full winner-takes-all crisis automation is deferred until false-positive and stability thresholds are met in shadow mode. - Snapback tests are retained for measurement; automatic Bayesian smoothing retuning from snapback output is deferred.

7. Phase 3: Strategic Executive (Trader Ensemble)

RL ensemble for trade execution decisions.

7.1 Ensemble Composition

Acceptance Criteria: - Ensemble includes SAC, PPO, and TD3 or justified alternatives. - Each policy has defined training cadence and evaluation metrics. - Optional market-making policy head is supported for low-volatility liquidity-provision windows, disabled by default until dedicated validation gates pass.

7.2 Decision Mechanism

Acceptance Criteria: - Decision policy uses a maximum entropy framework. - Multi-threshold genetic algorithm is specified for action selection. - Genetic algorithm evolution and search run offline in training and calibration. - Live execution uses teacher-student distillation, where the student policy mimics teacher decisions under strict latency budgets. - Student promotion requires teacher-student agreement thresholds on crisis slices, not only average-day performance. - Offline teacher monitoring remains active in production analytics, and student drift beyond threshold triggers automatic demotion. - Fast Loop inference is restricted to distilled student policy only; teacher inference is blocked from execution-critical path. - Student model artifacts require deterministic runtime profiles and approved compression plan (for example quantization/pruning/distillation) before promotion.

7.3 Observation Space

Acceptance Criteria: - Observation space includes outputs from Technical, Regime, Sentiment, and Consensus agents. - Features include regime probabilities, VaR and ES levels, macro differentials, and sentiment scores. - Observation schema is versioned and validated. - Fast Loop microstructure set explicitly includes order-book imbalance and queue-pressure features when feed quality is healthy.

7.4 Reward and Utility Functions

Acceptance Criteria: - Reward library includes RA-DRL composite, Sharpe, Sortino, Calmar, Omega, and Kelly variants. - Dynamic weighting is tied to regime state and sentiment signals. - Downside penalties increase during RBI outer band or negative sentiment spikes. - Formulas are documented in an appendix or code reference. - Liquidity-provision reward variant is defined for optional market-making mode, including maker-rebate capture net of adverse-selection and inventory-risk penalties.

7.5 Deliberation Process

Acceptance Criteria: - Critical-path thinker logic is lightweight and deterministic; heavy reasoning remains in the Slow Loop. - Simulations and policy search are removed from the execution-critical path and run asynchronously. - Execution consumes the latest validated policy snapshot, with scheduled refresh (e.g., every 10 minutes) or earlier on async completion. - Deliberation bypass is enabled under high-volatility triggers and does not block order decisions. - Trade rejection is tied to risk-control breaches, stale-data TTL breaches, or latency-cap breaches, not pending simulation completion. - Fast Loop peak-load decision stack targets p99 <= 8 ms as stretch, with enforced degrade or bypass safeguards above 10 ms for in-process compute stages. - Slow Loop latency (100 to 500 ms typical) is monitored separately and cannot directly bypass Fast Loop safety gates. - Release gates include jitter and tail-latency tests (p99.9) for decision-path determinism during peak-load simulations.

7.6 Safe RL Training and Deployment

Acceptance Criteria: - Online learning is gated behind offline training and paper trading validation. - Exploration is constrained by risk limits and hard safety filters. - Promotion to live trading requires risk committee approval and rollback readiness. - Optional online RL micro-updates (15 to 30 minute cadence) are allowed only in paper or shadow mode first, then promoted only after non-regression evidence and risk sign-off.

8. Portfolio Construction and Risk Budgeting

Dedicated allocation and exposure management. Acceptance Criteria: - Risk budgets per asset class and sector are defined. - Correlation and concentration limits are enforced. - Position sizing ties to forecast uncertainty and liquidity. - Inventory and exposure state model tracks gross, net, and hedge-adjusted risk even when strategy is directional. - Hedging policy engine is a first-class action type with cross-asset correlation checks and explicit risk-unwind logic. - Capacity ceiling is defined: initial live notional INR 5-10 Cr, with scale to INR 25 Cr only after 6 months of live validation; stress-tested at 1x, 2x, and 3x capacity, with a hard cap at 2x if impact exceeds 0.25 percent at 3x. - Max participation limits are documented and enforced at execution time. - Downside risk emphasis is enforced during panic regimes.

8.1 Market-Making Readiness Module (Dormant by Default)

Readiness module for future quoting strategies without changing current directional deployment scope. Acceptance Criteria: - Inventory controls are reusable for both directional and quoting modes. - Hedging action library includes partial unwind, cross-asset hedge, and emergency neutralization playbooks. - Enablement flag is default-off in production until dedicated market-making validation gates are passed.

9. Stress Testing Framework

Dedicated stress testing framework to validate resilience under adverse conditions. Acceptance Criteria: - A maintained scenario library exists with versioned definitions and expected outcomes. - Explicit shock scenarios include an RBI surprise rate hike, INR flash move, and liquidity drought. - Historical scenario set explicitly includes 2008 Global Financial Crisis data and 2013 Taper Tantrum regime shocks. - Stress tests cover historical crises and synthetic shocks at 1x, 2x, and 3x capacity. - Results are reviewed quarterly and trigger risk budget or capacity adjustments. - Impossible-scenario tests are mandatory, including correlation inversion, frozen constituent prices, and simultaneous multi-asset liquidity vacuum. - Data poisoning and feed-freeze simulations are included, with checks for correct behavior when no ticks or no book updates arrive for defined windows. - Validation confirms the engine distinguishes zero volume from missing data and enters safe mode when feed integrity is uncertain. - Snapback tests measure how many ticks it takes to clip net exposure after flash-shock conditions; smoothing parameters are capped if recovery is too slow. - Quote-response simulator tests are included to measure reaction speed, inventory drift, and unwind behavior under rapid microstructure changes.

10. Execution and Market Impact

Order routing and slippage management tailored to NSE. Acceptance Criteria: - Execution supports partial fills, pacing, and participation limits. - Slippage model is validated against intraday data. - Circuit breaker and halt handling is implemented and tested. - Execution metrics include fill rate, slippage vs model, and latency. - Market impact model is defined and linked to capacity assumptions. - Real-time routing health checks are enforced, with automatic degradation to reduce-only or close-only on execution infrastructure failure. - Real-time impact monitor tracks realized slippage against participation and ADV in near-real-time; position sizing is reduced automatically when impact breaches thresholds.

11. Phase 4: Independent Risk Overseer

Safety controls and oversight independent of trading logic. Acceptance Criteria: - Kill switch and loss limits enforced at broker and system levels. - Controls include max drawdown, daily loss, and position size constraints. - Extreme negative sentiment spikes trigger protective actions. - Automated drift detection uses ADDM or equivalent. - XAI logging captures top features and agent contributions using SHAP. - Kill switch hierarchy is documented with layered triggers (model, portfolio, broker, manual) and escalation order. - XAI coverage thresholds are defined (e.g., >= 80 percent of trades with top-k explanations logged). - Operating modes are explicitly defined as normal, reduce-only, close-only, and kill-switch with trigger and recovery criteria. - Dynamic risk budgets are volatility-aware; if realized volatility exceeds configured sigma thresholds, exposure caps auto-scale down before hard kill-switch triggers. - Data provenance reliability scoring dynamically adjusts position sizing instead of fixed haircut rules. - OOD or alien-state triggers enforce staged de-risking before full neutralization unless hard limits are breached. - Crisis taxonomy explicitly defines full crisis, agent divergence, and slow-crash or freeze states, each mapped to mode transitions and overrides. - Sentiment-to-price mismatch circuit breaker can downgrade sentiment to neutral during suspected manipulation or panic slang bursts. - Crisis entry requires multi-condition confirmation (volatility break, liquidity deterioration, and agent confidence threshold). - Crisis triggers must persist for configured ticks or seconds before activation, with explicit cooldown and hysteresis rules. - Crisis mode has max-duration expiry and auto-reverts unless revalidated by trigger conditions. - Agent divergence has explicit neutral-hold duration and staged re-risking rules after alignment recovery. - Risk overseer can trigger hedge or unwind actions directly when exposure or correlation thresholds breach limits. - Live PnL attribution is available per agent and per signal family to support real-time diagnosis during stress windows.

11.5 Phase 4.5: Hardware Acceleration and Low-Latency Track (Optional)

Targeted hardware evaluation path for execution-critical latency improvements. Acceptance Criteria: - Candidate components for acceleration are limited to execution-path critical steps (for example market data decode/normalization and selected pricing kernels). - Candidate components can include order-book parsing and routing adapters implemented in Rust/C++ microservices before FPGA escalation. - FPGA evaluation includes deterministic latency benchmarking: p50, p95, p99, p99.9, jitter, and worst-case tail behavior. - Hardware path has parity tests versus software baseline for correctness and failure handling. - Promotion requires measured latency/efficiency gain with no regression in risk controls, observability, or operational recovery. - Policy is enforced: non-critical workloads remain CPU/GPU to avoid unnecessary hardware complexity.

12. Phase 5: Validation and Evolution Loop

Backtesting, paper trading, and continuous learning. Acceptance Criteria: - Vectorbt or equivalent framework is used for realistic simulations. - Backtests span 2018 or 2019 to present, covering COVID crash, budget cycles, elections, and Oct-Nov 2022 volatility. - Walk-forward evaluation and time-based CV are required. - Leakage tests include timestamp audits and feature lag checks. - Survivorship bias controls are enforced using point-in-time universe data, including delisted symbols. - Nightly incremental learning uses TTM or equivalent lightweight models. - PEARL-based meta-learning adapts to the most recent 1 to 3 month regimes. - Training stability tests run via automated scripts like test_retraining.py. - Weekly sentiment re-fine-tuning is validated against SEntFiN or equivalent. - Synthetic data usage policy is documented, including labeling, scope boundaries for text and price data, and exclusion rules for performance claims unless validated on real data. - Stress library includes structural breaks and synthetic payment or policy shock scenarios with clear labels for synthetic-origin data. - Dual-loop boundary tests verify that text floods alone cannot flip portfolio direction without confirming technical or risk conditions. - Validation harness includes quote-response simulation slices for readiness testing of future market-making behaviors. - Phase 5 includes a dedicated research loop for rapid hypothesis testing (for example new z_t context variables) with explicit accept or reject criteria.

13. MLOps and Model Governance

Versioning, reproducibility, and controlled deployment. Acceptance Criteria: - Model registry includes data snapshot and code hash. - Champion-challenger promotion gates are codified. - Rollback policy exists and is tested. - Training pipelines are reproducible from raw data. - Human oversight gates are defined, including review cadence and approval authority. - Material strategy changes (latency bypass, dual-speed sentiment, novelty safeguards) require ablation evidence before promotion. - Distributed training posture is defined for scale-up (parallelized nightly TTM/PEARL jobs) with cost and reproducibility controls. - Research-to-production lead time is measured end-to-end (idea -> backtest -> paper -> shadow -> live) and reviewed at least monthly. - Experiment throughput is tracked (experiments per day/week) by pod and strategy family. - Compute cost per approved strategy update is tracked and budget-guarded. - Experiment failure taxonomy is reported (data-integrity failures vs model-logic failures) with corrective-action ownership. - Fixed uplift claims (for example fixed Sharpe, drawdown, accuracy, or speedup multipliers) are prohibited in approval artifacts unless backed by controlled benchmark or shadow A/B evidence. - Technology-change approvals (language/runtime/hardware) must report measured p99 and p99.9 deltas versus baseline plus correctness, observability, and recovery parity evidence.

14. Rollout Strategy and Live Monitoring

Controlled deployment with shadow testing and gating. Acceptance Criteria: - Shadow mode runs for a fixed minimum window. - Promotion requires challenger outperforming champion on risk-adjusted metrics. - Shadow A/B is mandatory for major strategy or model changes in USD/INR and gold tracks, and optional for lower-risk scope with risk approval. - Monitoring dashboards include PnL attribution, drift, and risk limits. - Promotion gates require no regression in Sharpe, drawdown, and slippage, plus acceptable rates for mode switching and false kill-switch events. - Crisis logic and override routing must complete shadow-mode validation before live activation. - Override rollout is sequential: enable one override at a time, validate, then promote the next. - Every production change must include expected impact statement, measured impact window, and rollback trigger before deployment approval. - Expected impact statement uses standardized evidence-first wording: unknown until controlled benchmark or shadow A/B; target is non-regression plus measurable improvement. - Every change request must state go criterion as all applicable phase-gate checks passing. - Every change request must state no-go criterion as any latency, risk, or correctness gate failure. - Every promotion request must include completed governance templates for release record and CI benchmark evidence checklist. - Mandatory 48-hour post-deploy review covers latency, slippage, risk events, drift, and incident count.

15. Operations, Security, and Resilience

Reliability, incident response, and secure operations. Acceptance Criteria: - Secrets management is centralized and audited. - DR plan defines RPO and RTO targets. - Centralized logging with retention policies is implemented. - Incident response runbook exists and is exercised. - System time sync is enforced via NTP or equivalent with drift alerts. - Observability includes metrics, traces, and alerts for data latency, inference time, and execution failures. - Operational dashboards track decision staleness, feature lag, mode-switch frequency, OOD trigger rate, kill-switch false positives, and MTTR. - Cross-functional pods are the default operating model: Quant + Engineer + Trader or Risk + Data per asset cluster. - Pods own end-to-end production metrics (slippage, stability, uptime, tail-risk incidents) and run a shared production incident rotation.

16. Milestones and Deliverables

Phased execution with measurable outcomes. Acceptance Criteria: - Each phase has exit criteria, owners, and timelines. - Go-live criteria include performance, stability, and compliance readiness. - Every milestone artifact has pod-level ownership with a primary DRI and backup reviewer.

16.1 Phase Go or No-Go Benchmark Gates

Phase | GO Benchmarks (all required) | NO-GO Trigger

Phase 1: Data Orchestration | Data uptime >= 99.5 percent during NSE hours for at least 10 consecutive trading days; core symbol completeness >= 99.0 percent; provenance tagging coverage = 100 percent; leakage tests = 100 percent pass | Any critical data integrity failure unresolved, or any required benchmark not met

Phase 2: Analyst Board | Baseline and ablation packs are complete for each active model family; sentiment precision/recall thresholds and timestamp alignment tests pass; OOD and regime transition validations pass documented thresholds | Missing baseline evidence, failed sentiment/OOD checks, or unresolved model-data alignment defects

Phase 3: Strategic Executive | Fast Loop decision stack meets latency release gates (stretch p99 <= 8 ms, enforced degrade path above 10 ms, and p99.9 jitter limits); teacher-student agreement threshold passes on crisis slices; rollback path is tested | Latency gate breach without compliant degrade behavior, failed crisis-slice agreement, or untested rollback

Phase 4: Independent Risk Overseer | Kill-switch hierarchy drills pass (model, portfolio, broker, manual); dynamic volatility-scaled budget controls function in stress drills; hedge and unwind triggers behave as specified | Any failed kill-switch drill, risk limit breach without protective transition, or unresolved false-trigger escalation path

Phase 4.5: Hardware Acceleration (Optional) | Candidate component shows pre-defined benchmark improvement versus software baseline on p99 and p99.9; correctness parity and failover parity are validated; observability and incident recovery remain compliant | No measured latency benefit, any correctness mismatch, or degraded recovery and observability

Phase 5: Validation and Evolution | Walk-forward, time-based CV, leakage, and dual-loop boundary tests pass; shadow A/B non-regression is confirmed for risk-adjusted metrics and slippage; stress and impossible-scenario library passes signed review | Any validation failure, statistically material regression in shadow A/B, or unreviewed stress-test exceptions

Phase 17: Gold Expansion | Core stack has already passed go-live checklist; COMEX shock scenario limits and gold-specific notional/participation controls pass drills; gold shadow window meets risk and stability gates | Base stack not go-live eligible, failed gold stress controls, or unresolved gold-specific execution risk defects

Go-Live Checklist (Minimums): - Minimum 3 calendar months of paper trading with >= 80 percent uptime. - Annualized Sharpe >= 1.8. - Sortino >= 2.0. - Max Drawdown <= 8 percent. - Win rate >= 52 percent. - Profit factor >= 1.5. - Average realized slippage <= model estimate + 20 bps. - Data uptime >= 99.5 percent during NSE trading hours. - Zero critical compliance violations and zero broker or SEBI flags or order rejections.

17. Gold Expansion (MCX Bullion Track)

Add exchange-traded gold alongside existing USD/INR and NSE instruments using the same multi-agent stack. Scope and Assumptions: - Instruments: MCX GOLD and GOLDM futures in Phase 1; options only after stable futures rollout. - Venue: Indian exchange-traded commodity segment only. - Dependency: Existing core stack has passed current go-live checklist. - Gold risk policy includes scenario-based notional and participation limits during COMEX-linked shock windows.

18. Next Iteration Enhancements (Nice-to-have)

Planned upgrades to be prioritized after initial production stability. Acceptance Criteria: - Backlog is reviewed quarterly and scoped only after six months of stable live operation. - Any new enhancements require documented risk review and rollback plans before deployment.

Prioritization Framework: - High-impact / low-effort: order-book imbalance reinforcement in Fast Loop, real-time slippage-impact controls, dynamic volatility-scaled risk budgets. - Medium-impact / medium-effort: optional market-making mode, Rust/C++ critical-path microservices, live PnL attribution dashboard. - High-impact / high-effort: FPGA path for select components, full online RL fine-tuning in production.

18.1 Evidence-First Prioritization Table

Priority | Initiative | Effort | Expected Impact (Hypothesis Only) | Go Criterion | No-Go Criterion

1 | Order-book imbalance reinforcement in Fast Loop | Low to medium | Improved short-horizon microstructure response without latency regression | Fast Loop p99/p99.9 non-regression versus baseline and statistically valid shadow non-regression on slippage/risk metrics | Latency regression, data-quality instability, or statistically material shadow regression

2 | Real-time slippage-impact monitor with dynamic volatility-scaled risk budgets | Medium | Reduced adverse execution in stressed windows and lower tail exposure | Trigger logic passes replay and live-shadow drills; stress-window slippage and risk incidents improve versus baseline without return degradation beyond gate limits | Missed breach handling, excessive false positives, or risk-limit control failure

3 | Dedicated research loop (Phase 5 hypothesis cycle) | Medium | Faster validated iteration and cleaner research-to-production throughput | Weekly hypothesis pipeline, experiment taxonomy, and promotion evidence complete; lead-time KPI improves without governance regressions | Repeated unclosed hypotheses, high unresolved data-integrity failure ratio, or missing promotion evidence

4 | One Rust/C++ critical-path microservice (for example parser or routing adapter) | Medium to high | Lower tail latency and jitter on a targeted execution component | Predefined benchmark gain on p99 and p99.9 versus Python baseline plus correctness/failover/observability parity | No measurable gain, correctness mismatches, or degraded incident recovery

5 | FPGA or advanced hardware acceleration path | High | Additional execution-path tail-latency reduction for proven bottlenecks | Phase 4.5 gates pass with measured latency gain, correctness parity, and operational recovery parity | No benchmark advantage, elevated complexity without risk-adjusted benefit, or degraded resilience

Appendix A: Metrics and Promotion Gates

Acceptance Criteria: - Core metrics include Sharpe, Sortino, Calmar, MDD, turnover, slippage, and capacity. - Promotion gates include statistical significance thresholds vs baseline. - Risk budget adherence is validated before production promotion.

Appendix B: Data Quality SLAs

Acceptance Criteria: - Feed uptime thresholds are defined and monitored. - Max permissible latency is defined per data source. - Data anomaly alerts trigger automatic fallback or pause.

Appendix C: Indian Market Optimization Summary

Area | Optimization

Data and Macros | RBI policy, FII/DII flows, domestic inflation metrics, FX reserves, and sentiment sources

Regime Detection | RBI inner and outer band monitoring with event calendar tracking

Volatility Modeling | GARCH quantiles tuned for NSE circuit breakers and FII movement patterns

Reward Emphasis | Sortino and Calmar weighting during panic regimes

Consensus Signal | Bayesian LSTAR or ESTAR hybrid with sentiment integration

Trading Execution | SAC-led ensemble for regime-switch robustness

Infrastructure | Mumbai VPS deployment with static IP whitelisting

Hardware Path | Optional FPGA acceleration for execution-critical components only

Operating Model | Cross-functional pods with shared outcome and incident accountability

Latency Discipline | Fast Loop stretch p99 <= 8 ms with mandatory degrade/bypass above 10 ms